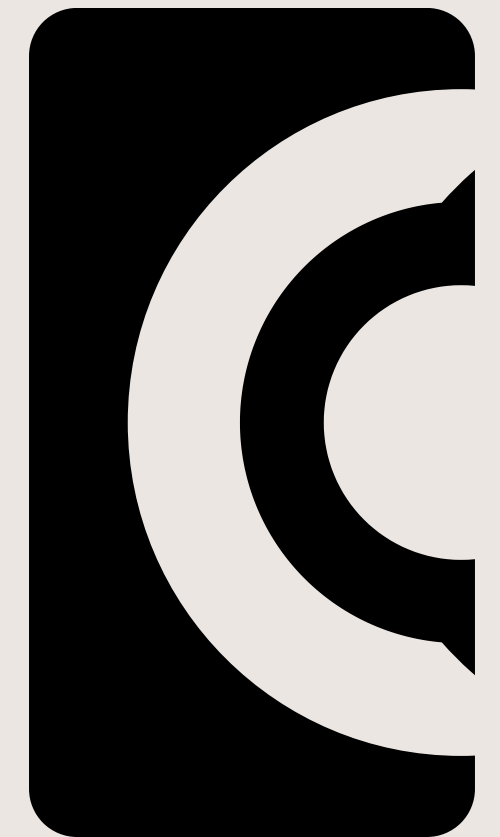


Iteration 2

EasySpot

Parking Analytics Platform



agenda

Architecture Notebook

Prototypes for the core user
stories

Key quality requirements

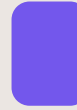
The primary focus of EasySpot is to enable a system that has a low latency data flow for real-time occupancy events and a persistent and scalable database that stores historical and statistical data.

Non-functional requirements and constraints

 **Performance and Responsiveness**

 **Scalability**

 **Data Consistency and Integrity**

 **Deployment and Portability (Constraints)**

Key quality requirements

The primary focus of EasySpot is to enable a system that has a low latency data flow for real-time occupancy events and a persistent and scalable database that stores historical and statistical data.

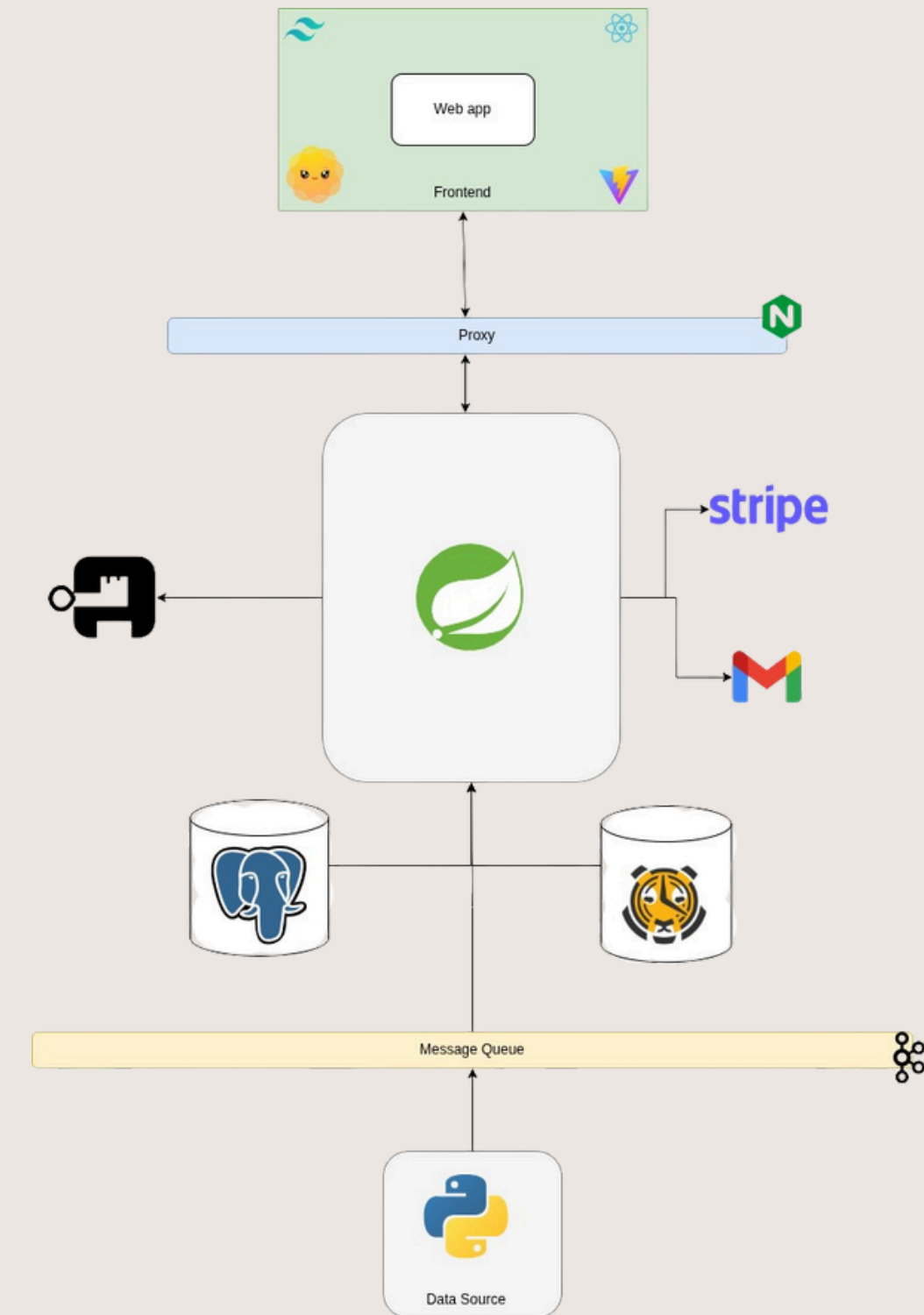
Functional requirements

-  Real-time Monitoring & Alerts
-  Parking Simulation

Layered Architecture

System components

EasySpot is structured into four main components: Data Sources, Message Queue, Backend, and Frontend. These components work together to ensure a smooth flow of data from generation to user interaction, enabling real-time processing and efficient system performance.



Data Flow & User

Interaction

Data & Processing

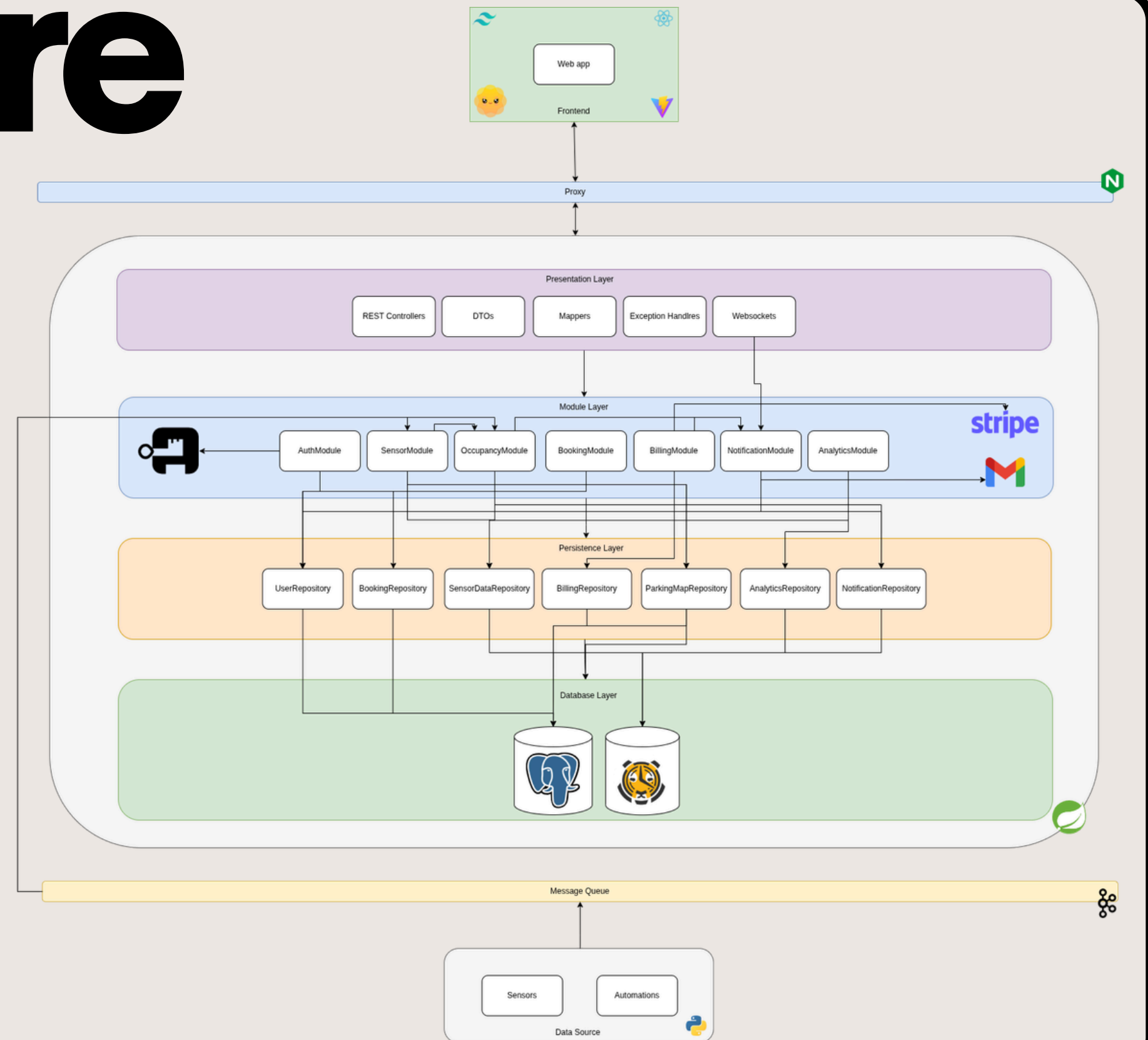
The system uses simulated sensors built in Python to generate real-time data. This data is streamed through Apache Kafka and processed by the backend using Spring Boot. The system stores information in PostgreSQL for relational data and TimescaleDB for time-series data.

User Side

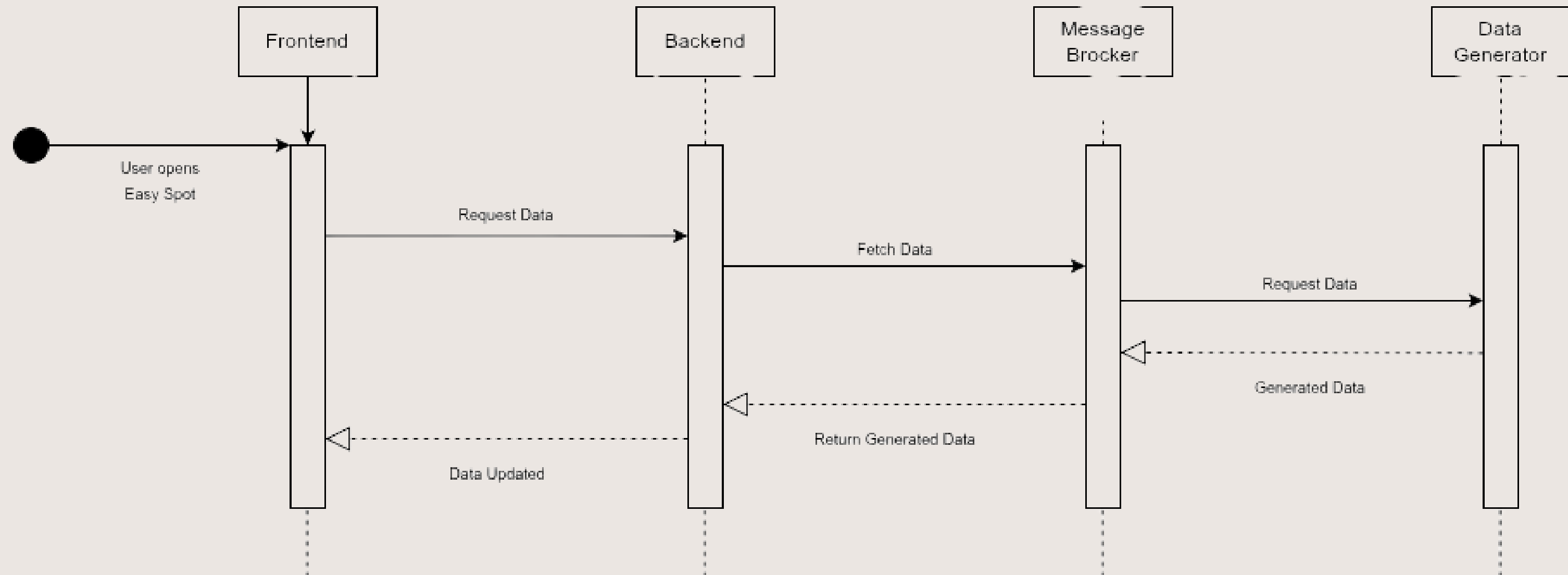
The user interacts with a web application built using React. The app communicates with the backend through REST APIs for actions like searching and booking, and uses WebSockets to receive real-time updates. This allows users to view available parking spots and live changes instantly.

Architecture

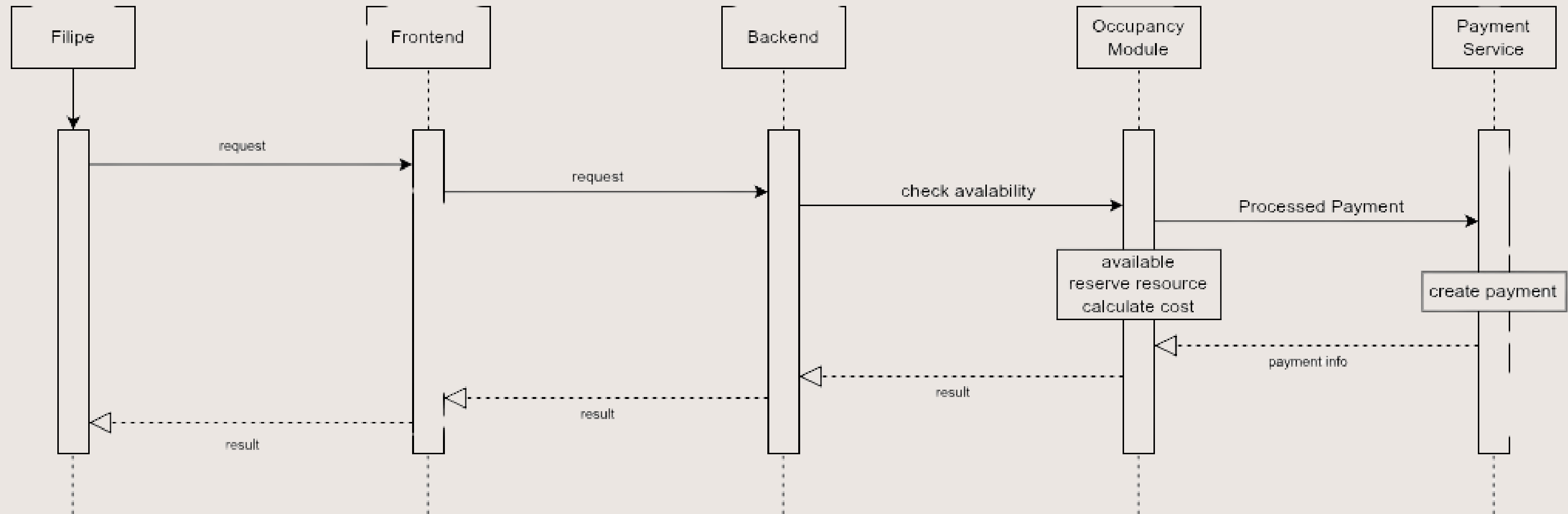
Our architecture is divided into independent modules within the logic layer. Each module communicates with its respective repository, but we also allow for inter-module communication to handle complex business workflows.



User Open Easy Spot



US#3 Spot Reservation



Prototypes

Prototype: Driver

User Story #1 - #11

Lista de Parques

Mapa

Custos

Favoritos

Perfil

Estacionamento

Pesquisar parque ou zona...

Todos os lugares ▾

Com Lugares

Carregador EV

Acessível

12 parques encontrados

DISPONÍVEL   Estádio Cidade de Coimbra /HORA Rua Dom Manuel I, Coimbra 38% LIVRES 46/120 9 min	DISPONÍVEL  CoimbraShopping /HORA Av. Mendes Silva, Coimbra 28% LIVRES 17/60 30 min	DISPONÍVEL   Fórum Aveiro /HORA R. do Batalhão de Caçadores 10, Aveiro 26% LIVRES 22/85 5 min	DISPONÍVEL   Glicínias Plaza /HORA R. Eng. Von Haff, Aveiro 33% LIVRES 50/150 22 min
QUASE CHEIO  Europa /HORA Avenida Marquês de Pombal, Leiria 10% LIVRES 9/90 8 min	DISPONÍVEL   Estádio Municipal Dr... /HORA Caminho da Ribeira, Leiria 40% LIVRES 120/300 18 min	DISPONÍVEL  Gaivotas /HORA Av. do Mar, Figueira da Foz 43% LIVRES 30/70 10 min	DISPONÍVEL   Foz Plaza /HORA Rua dos Condados, Buarcos, Figueira da Foz 31% LIVRES 78/250 28 min
QUASE CHEIO Estação Ferroviária de Ovar /HORA Largo da Estação, Ovar 21% LIVRES 10/48 4 min	DISPONÍVEL   Praia do Furadouro /HORA Av. Central, Furadouro 83% LIVRES 100/120 17 min	DISPONÍVEL   Mercado Municipal de... /HORA Av. José Augusto de Carvalho, Arganil 32% LIVRES 19/60 3 min	DISPONÍVEL  Montalto /HORA R. do Montalto, Arganil 94% LIVRES 94/100 32 min

Prototype: Maintenance Technician

User Story #13: Infrastructur Mapping & Status Monitoring

User Story #14: Repair & Update of sensors

PAINEL TÉCNICO

 Painel Técnico

 Diagnóstico & Manutenção

Mapa

Definições

Painel Técnico

Segunda-feira, 9 de março de 2026 · Laura Farias



33

Total Sensores

— 27 operacionais



96.4%

Uptime Médio

▲ 5 com problemas



5

Falhas Ativas

4 falha · 1 offline

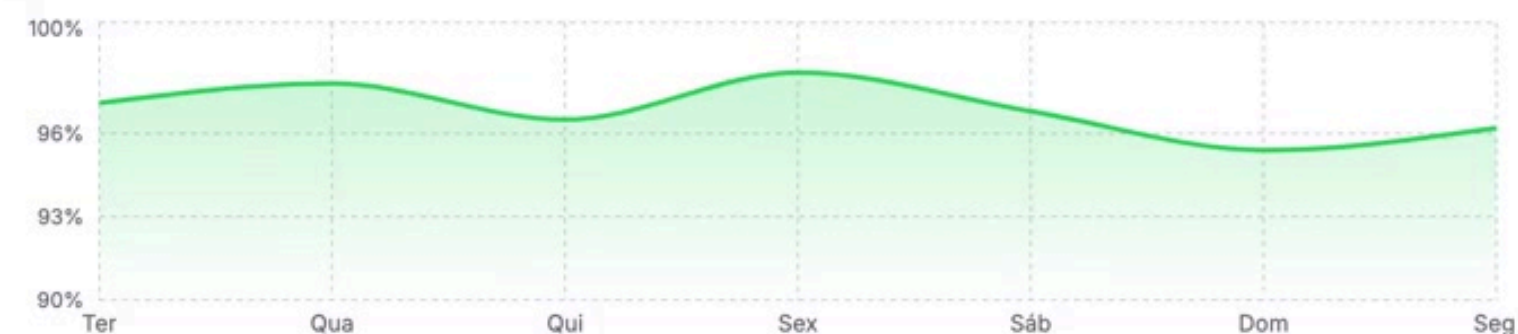


4.2h

MTTR

~ Tx. F.Pos: 3.9%

Uptime Global — Últimos 7 Dias



Estado dos Sensores



- Operacional

27 82%

4 12%

● Offline

1 3%

- Manutenção

1 3%

⚠ Ordens Urgentes

4 pendent



Substituição emitter IR – Lugar B7

Critica

Fórum Aveiro · IR-AV1-B07 · Prazo: 09/03, 12:00

Atualizar



Restauro de comunicação – RFID Entrada

Critica

Europa – Leiria · RFID-LE1-ENT1 · Prazo: 09/03, 11:00

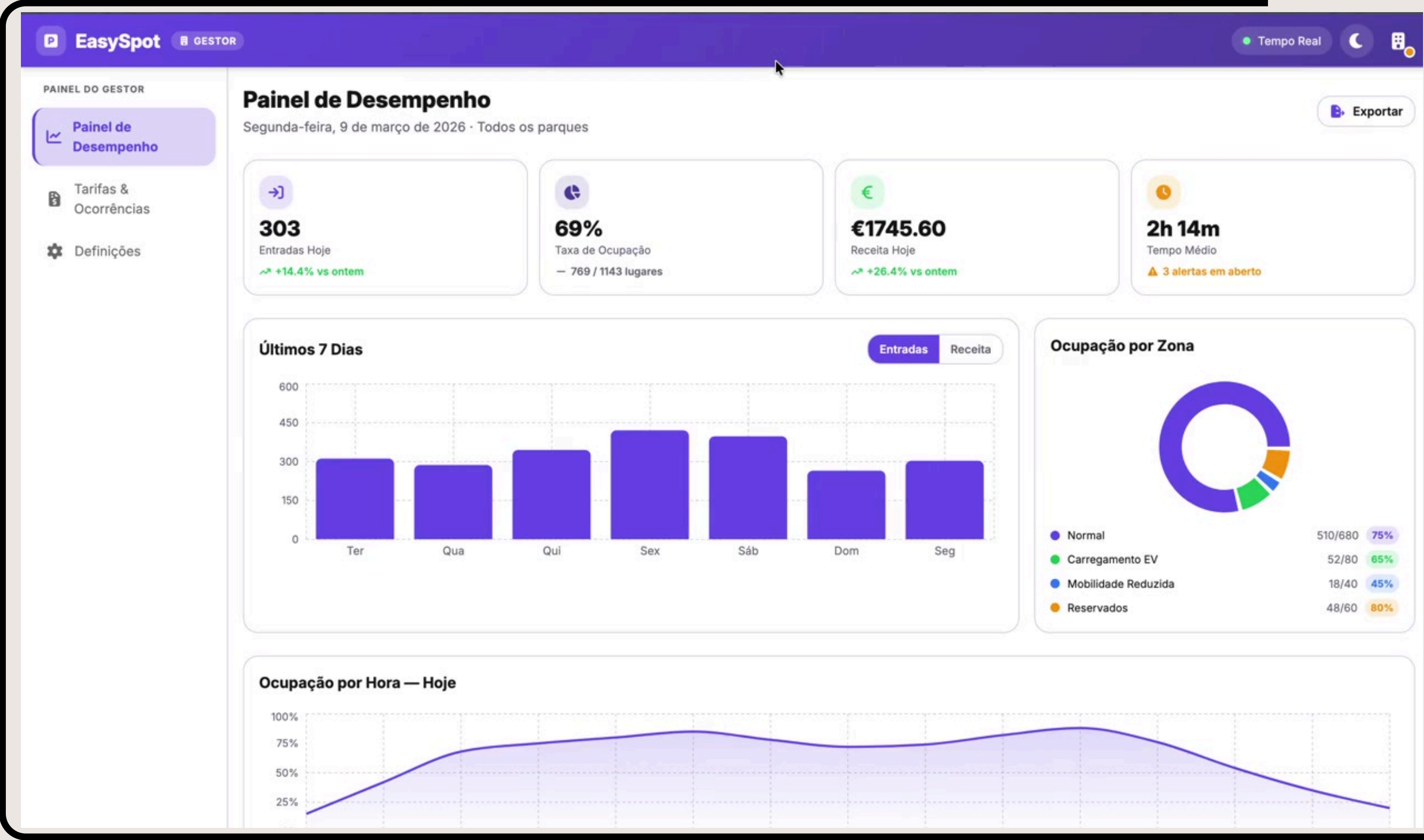
Atualizar

Prototype:

Parking Operations Manager

User Story #10: Real-Time Occupancy & Financial Performance

User Story #11: Tariff Management & Issue Oversight



Painel de Desempenho

Segunda-feira, 9 de março de 2026 · Todos os parques

Exportar



303

Entradas Hoje

+14.4% vs ontem



69%

Taxa de Ocupação

769 / 1143 lugares



€1745.60

Receita Hoje

+26.4% vs ontem



2h 14m

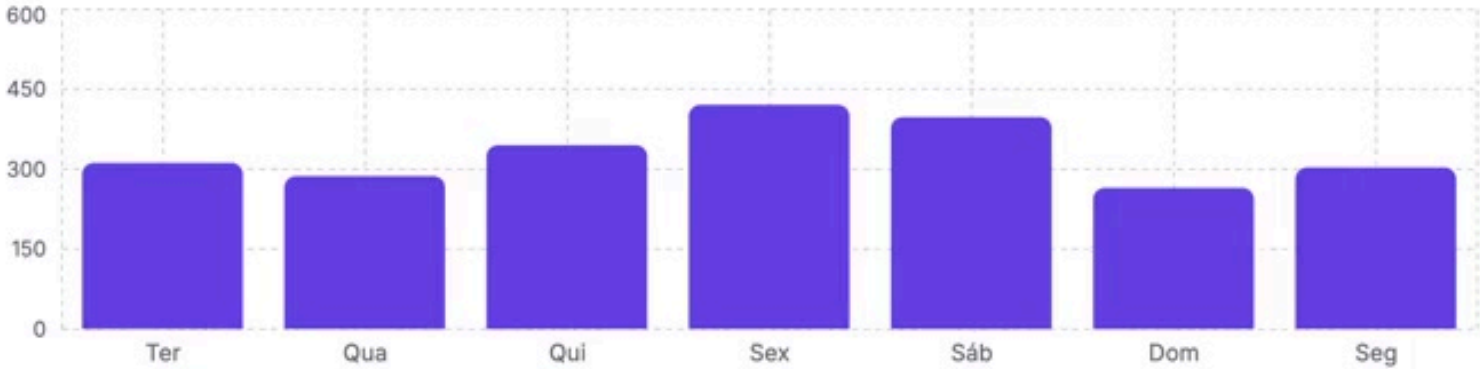
Tempo Médio

3 alertas em aberto

Últimos 7 Dias

Entradas

Receita



Ocupação por Zona



Normal	510/680	75%
Carregamento EV	52/80	65%
Mobilidade Reduzida	18/40	45%
Reservados	48/60	80%

Ocupação por Hora — Hoje



thank you

EasySpot

